



ST BENEDICT'S

SUBJECT	PHYSICAL SCIENCE: CHEMISTRY
GRADE	11
EXAMINER	Mr Ntombela
MODERATOR	Mrs Govender
DATE	31 July 2024
MARKS	135
DURATION	2 hours

%

LEARNER NAME: MEMO

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 20 pages and a separate data sheet. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Additional space is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	Total
Mark										
Moderated Mark										
Question Total	12	27	9	17	15	14	18	8	15	135

Comment:

QUESTION 1**GR****12 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	C	D
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Here the option C has been marked as an example.

1.1	A	B	C✓✓	D
1.2	A	B	C✓✓	D
1.3	A	B	C	D✓✓
1.4	A	B	C✓✓	D
1.5	A	B	C✓✓	D
1.6	A	B	C	D✓✓

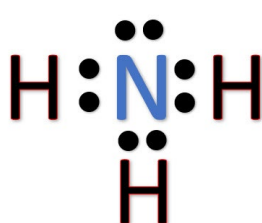
QUESTION 2 CHEMICAL BONDING KH**27 MARKS**2.1 Define the term *polar covalent bond*. (2)

A covalent bond that has an (unequal sharing of electrons) leading to a dipole forming ✓✓
 Has an electronegativity difference..

2.2 Determine the bond polarity of ammonia. (3)

Electronegativity difference = $3,0 - 2,1 = 0,9$ ✓
 END greater than zero ✓, polar covalent bonds ✓
 (Unequal sharing of electrons)

2.3 Draw the Lewis diagram of ammonia. (3)



Valence electrons

Nitrogen ✓

Valence electrons for

Hydrogen ✓

Correct bonding ✓

2.4 Determine the molecule polarity of ammonia. (3)

Ammonia has electron cloud asymmetry ✓ due to its pyramidal shape ✓
 Ammonia is therefore a polar molecule ✓

2.5 Indicate the molecule polarity on your Lewis diagram. ✓δ- on N and ✓δ+ on H (2)

2.6 Name the type of intermolecular forces that exists between **ammonia** molecules. (1)

Hydrogen bond forces ✓

2.7 With reference to intermolecular forces, compare the boiling points of water and ammonia and fully explain this difference. (4)

- Both water and ammonia have polar molecules and therefore have **hydrogen bonds** between molecules. ✓
- Water forms 4 **hydrogen bonds** ✓ per molecule and ammonia forms 2 ✓ **hydrogen bonds** per molecule
- More energy** is required to **overcome** the greater number of hydrogen bonds
 Water has a higher melting/ boiling point. ✓

2.8 Define the term *ionic bonding*. (2)

Ionic bonding involves a **transfer of electrons** to form cations and anions ✓ which are electrostatically attracted in an ionic crystal lattice ✓

2.9 Write the half reactions for copper and chlorine. Label the reaction as oxidation or reduction. (4)

Copper: $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$ ✓ **Oxidation** ✓

Chlorine: $\text{Cl} + \text{e}^- \rightarrow \text{Cl}^-$ ✓ **Reduction** ✓

2.10 Which compound from the table above is a good conductor of electricity in a liquid phase? Explain your answer. (3)

Copper chloride ✓. The ions are not in a lattice anymore so can move past each other ✓ and charge can flow. ✓

QUESTION 3 ENERGY CHANGE KM**9 MARKS**

3.1 Define the term *activation energy*. (2)

The minimum energy✓ required for a reaction to take place✓

3.2 Does this energy profile diagram represent **EXOTHERMIC**✓ or **ENDOTHERMIC** reaction? Circle the correct answer. (1)

3.3 Explain your answer in 3.2. (2)

More energy is released than absorbed✓✓ OR energy of the products is less than energy of the reactants✓✓.

3.4 Calculate the heat of reaction. (2)

$$\Delta H = 183,3 - 400✓ = - 216,7 \text{ KJ.mol}^{-1}✓$$

3.5 Indicate on the energy profile above with a dotted line the effect of a catalyst on the activation energy. **Lower activated complex**✓ (2)

QUESTION 4 REACTION RATES FMC**17 MARKS**

4.1 The rates of the reactions are measured by mass lost per unit time.

Describe a precaution when using this method.

(2)

Place the cotton wool✓, so that ONLY the gas can escape the vessel✓

Or

Can only be used for heavy gasses as light masses will not change the mass on the scale.

4.2 In Experiment 1, in which flask, **W** or **X**✓, will the reaction rate be faster? Circle the correct answer.

(1)

4.3 Explain your answer in 4.2.

(3)

The smaller particle size, the greater the surface area exposed to the acid✓.

More collision per unit time. ✓

There will be more effective collisions per unit time✓. Therefore, faster reactions.

4.4 If the number of moles for hydrochloric acid is 0,1 moles, Calculate the volume of $\text{CO}_2(\text{g})$ produced at STP in flask **W**.

(3)

0, 1 : 0,05✓

$V = n \times V_m = 0,05 \times 22,4 = 1,12 \text{ dm}^3$ ✓

4.5 Consider the reaction conditions given for Experiment 2.

4.5.1 How will the rate of reaction in flask **Z** compare to that in flask **Y**? Choose from GREATER THAN Y, LESS THAN Y or **EQUAL TO Y**✓.

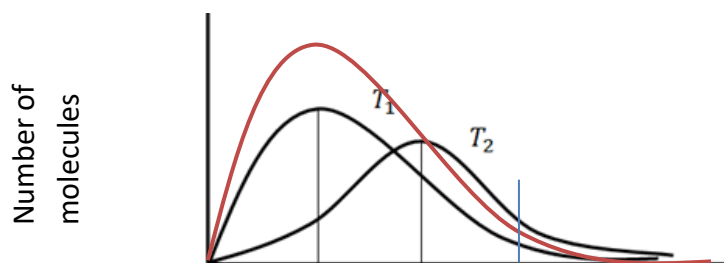
(1)

4.5.2 It is observed that the total volume of gas produced in flask **Z** is greater than that produced in flask **Y**. Account for this observation.

(2)

HCl is the limiting reagent✓, so there is now more of it to react. ✓.

4.6



4.6.1 Which graph represents a decrease in Temperature? **T1**✓ or **T2**.

Circle your answer.

(1)

4.6.2 **Line✓ and shaded area ✓**

(2)

4.6.3 .

(2)

Graph must be higher than **T1**✓✓

QUESTION 5 REACTION RATES FMC**15 MARKS**

- 5.1 Which TWO conditions have to be met for an effective collision to occur between colliding reactant particles? (2)

Correct orientation ✓

Sufficient kinetic energy to overcome the activation energy ✓

- 5.2 Use the collision theory to explain how an increase in the concentration of HCl would affect the rate of this reaction. (4)

Increasing concentration means that there is greater number of particles per unit volume. ✓

More collision per unit time. ✓

More effective collision per unit time. ✓

This increases the reaction rate. ✓

- 5.3 Describe the method used to collect the gas. (2)

Downward displacement of water ✓ – Invert a measuring cylinder filled with water ✓ into a water bath.

- 5.4 Shallower gradient ✓ and x 1,5 higher end ✓ (2)

- 5.5 A 2,1 g sample of nickel reacts with hydrochloric acid in a reaction vessel. 0,03 moles of hydrogen gas is collected. Assume the reaction to be 100% efficient. Calculate the percentage purity of the sample of nickel. (5)

0,03 ✓ (Ratio 1: 1)

$$= \frac{m}{59} ✓$$

$$m = 1,77 \text{ g} ✓$$

$$\begin{aligned} \% \text{purity} &= \frac{\text{Pure mass}}{\text{Impure mass}} ✓ \\ &= \frac{1,77}{2,1} \times 100 = 84,29\% ✓ \end{aligned}$$

QUESTION 6 STOICHIOMETRY SG**14 MARKS**6.1 Define the term *molar mass*. (2)

Mass in grams of one mole of that substance. ✓✓

6.2 Prove that the silver nitrate is the limiting reactant. (6)

 MgCl_2

$$n = c \cdot V = (0,25)(0,055) = 0,01375 \text{ mol}$$

 AgNO_3

$$n = c \cdot V = (0,15)(0,070) = 0,0105 \text{ mol}$$

 $\text{MgCl}_2 : \text{AgNO}_3$

1 : 2

$$n = \frac{0,0105}{2} = 5,25 \times 10^{-3} \text{ of } \text{MgCl}_2 \text{ required}$$

If AgNO_3 is limiting then $5,25 \times 10^{-3} \text{ mol}$ of MgCl_2 required. 0,01375 available (enough). AgNO_3 limiting. ✓

6.3 Hence, calculate the mass of silver chloride that can be formed during this reaction. (3)

 $2\text{AgNO}_3 : 2\text{AgCl(s)}$

2 : 2 ✓

$$n = 0,0105 \text{ mol}$$

$$n = \frac{m}{M}$$

$$0,0105 = \frac{m}{143,5}$$

$$m = 1,51 \text{ g}$$

6.4 Calculate the percentage yield of silver chloride if 1,49 g of silver chloride was in fact recovered. (3)

$$\% \text{yield} = \frac{\text{actual}}{\text{theoretical}} = \frac{1,49}{1,51} \times 100 = 98,98\%$$

QUESTION 7 STOICHIOMETRY SG**18 MARKS**

Consider the following two aqueous solutions:

7.1 Give a reason why the above solutions are standard solutions. (1)

They have known concentration. ✓

7.2 Define the term *solvent*. (1)

The substance in which another substance is dissolved forming a solution. ✓

7.3 Identify the solvent used in the above solutions. (1)

Water ✓

7.4 Give the method for making a standard solution using four key bullet points. (4)

Mass the required solid. ✓

Transfer all solute to the volumetric flask. ✓

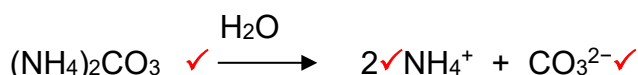
Rinse with distilled water. ✓

Make up the final volume with distilled water. ✓

The balanced dissociation reaction for ammonium hydroxide is given as:



7.5 Write balanced chemical equation showing the dissociation of ammonium carbonate. (3)



7.6 Solution A is now mixed with Solution B in a new bigger beaker. Calculate the new concentration of the ammonium ions in the solution. (5)

$$\text{NH}_4\text{OH}: n = c \cdot V = 0,2 \times 0,25 = 0,05 \text{ mol} \quad \checkmark$$

$$(\text{NH}_4)_2\text{CO}_3: n = c \cdot V = 0,3 \times 0,5 = 0,15 \text{ mol} \quad \times 2 = 0,3 \text{ mol} \quad \checkmark$$

$$c = \frac{n}{V} \quad c = \frac{0,05 + 0,3}{0,25 + 0,50} = 0,47 \text{ mol} \cdot \text{dm}^{-3}$$

7.7 Determine the volume of water that must be added to Solution B to dilute the solution to $0,03 \text{ mol} \cdot \text{dm}^{-3}$. (3)

$$n = c \cdot V = 0,3 \times 0,5 = 0,15 \text{ mol}$$

$$v = \frac{n}{c} = \frac{0,15}{0,03} = 5 \text{ dm}^3$$

$$5 - 0,5 = 4,5 \text{ dm}^3$$

QUESTION 8 STOICHIOMETRY KM**8 MARKS**

A student performed an experiment to determine the percentage of water of crystallization in hydrated sodium carbonate.

8.1 Explain what is meant by the term 'water of crystallization'. (2)

Water that is associated with crystal formation ✓✓

OR . "water of crystallization" refers to water that is found in the crystalline ✓
framework of a salt. ✓

The water of crystallization is normally expressed in the chemical formula of the substance. The formula for hydrated magnesium sulphate is **MgSO₄.xH₂O**.

8.2 Explain what is meant by the 'x' in the formula. (2)

number of water molecules ✓ in the crystal per unit of the salt ✓

8.3

8.3.1 Determine the number of moles of dehydrated Epsom salts (1)
0,01 ✓

8.3.2 Determine the number of moles of water (1)
0,07 ✓

8.3.3 Determine **x** in the formula: **MgSO₄.xH₂O** (2)

Ration is 0,01:0,07

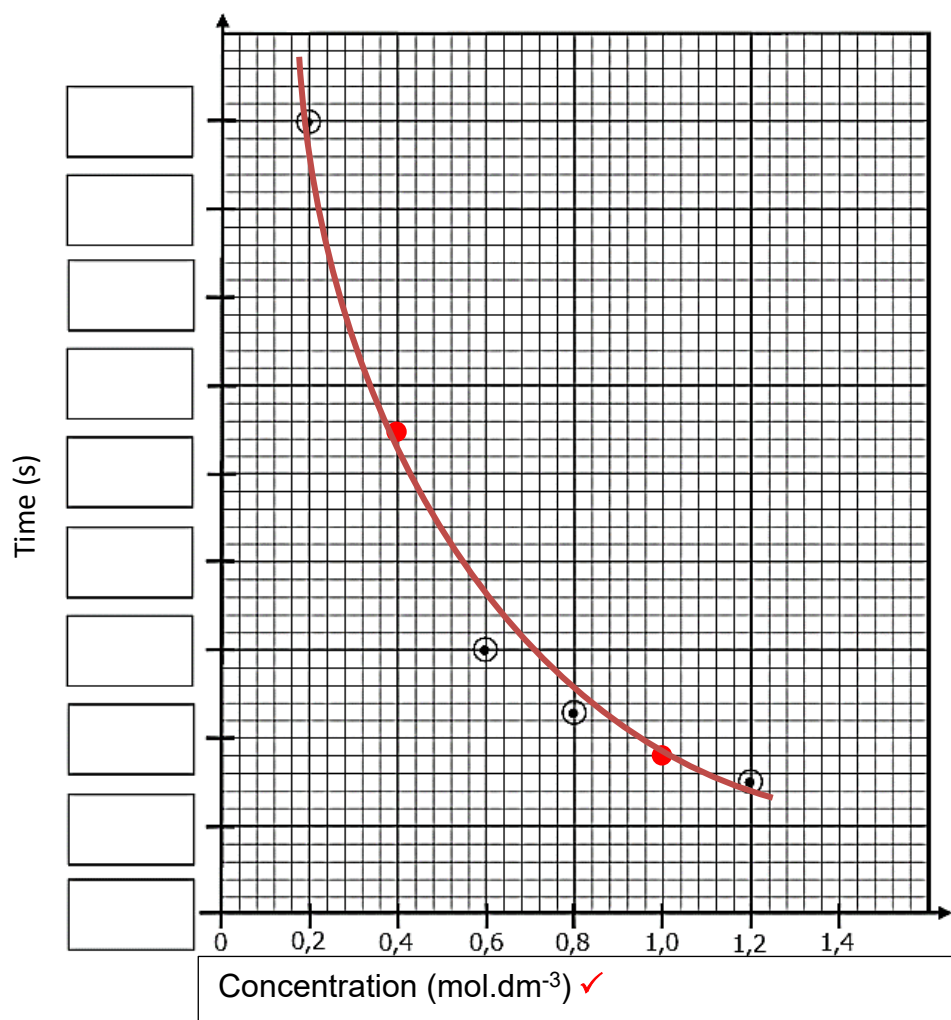
1:7 ✓

x = 7 ✓ (MgSO₄.7 H₂O)

QUESTION 9 SCIENTIFIC SKILLS ZN**15 MARKS**

- 9.1 Why will there be a decrease in the intensity of the light passing through the solution as the reaction proceeds? (1)

The solution is turning cloudy ✓



- 9.2 Write down a suitable label for the x-axis in the empty block provided. (1)
- 9.3 Determine the scale used on the y-axis and fill in the numbers in the empty blocks provided. **0, 20, 40, 60, 80, 100, 120, 140, 160, 180** ✓ (1)
- 9.4 Plot the points for experiments 2 and 5. (2)
- 9.5 Draw a best-fit line through the points. **Smooth curve** ✓✓ (2)
- 9.6 Write the independent variable for this experiment. (2)

Concentration ✓ of solution ✓

9.7 Reaction rate is directly proportional to concentration.

or

The time taken for the cross to disappear is directly proportional to the concentration.

9.8 Comment on the reliability for this experiment. (2)

Unreliable because the experiment was not repeated 3 time✓ **to get an average**

and

Good as more than 6 points were plotted. ✓

9.9 Comment on the accuracy of experiment 5 if it actually takes 34 s for the light intensity to decrease. (2)

$$\% \text{ accuracy} = \frac{\text{actual}}{\text{theoretical}} = \frac{34}{36} \times 100 = 94,44\% \checkmark \text{ only 1 mark}$$

$$\% \text{ accuracy} = \frac{|\text{actual} - \text{theoretical}|}{\text{theoretical}} \times 100 \checkmark$$

$$= \frac{34}{36} \times 100$$

$$= 5,56\%$$

Accurate to within 5,56%✓